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Challenges of spatio-temporal trajectory datasets

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ABSTRACT
The rapid evolution of wireless location technologies has significantly increased the availability of mobility data, particularly spatio-temporal trajectory data, detailing object movements and locations over time. This data holds diverse applications such as predicting travel patterns, optimizing routes, analysing social interactions, and managing urban resources. However, leveraging trajectory datasets presents challenges, including data heterogeneity, volume, velocity requiring real-time processing, privacy concerns with sensitive information, and regulatory compliance. Traditional systematic reviews of literature may not fully capture industry challenges, as professionals often do not publish research. To address this gap, industry expert input is vital. Collaborating with academia, we propose using focus group discussions to explore these challenges comprehensively. Focus groups offer a qualitative platform for professionals and academics to identify practical issues and gaps in the literature. This combined approach will inform technological advancements in creating and utilizing spatio-temporal datasets. Additionally, integrating insights from focus groups with a literature review ensures a thorough analysis of data challenges. This methodology aims to provide a holistic understanding and foster innovation in managing and exploiting trajectory data effectively.

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Data ethics; focus group study; privacy concerns; spatio-temporal datasets

1. Introduction

The rapid advancement in collecting spatio-temporal trajectory data has been driven by a convergence of various factors. These factors encompass the widespread embrace of Global Positioning System (GPS)-enabled smartphones, increased investment in sustainable transportation and smart city technology, and the proliferation of the Internet of Things (IoT) and sensor-connected devices (Yan 2011). As a result, sophisticated techniques and tools have surfaced to examine spatio-temporal data across a broad spectrum of research applications. These applications encompass real-time traffic control, urban planning, environmental surveillance, marketing, advertising, analysis of human behaviour and activities, disaster

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management and response, optimisation of telecommunication networks, and ensuring individual health and safety (Arslan, Cruz, and Ginhac 2019b; Zheng 2015).

Despite the numerous advantages linked to trajectory datasets, obstacles continue to exist in their application. These obstacles encompass unresolved queries and issues concerning dataset creation, preprocessing methods, data confidentiality, strategies for secure dataset sharing to prevent data breaches, and the utilisation of simulated and confidential datasets. To attain a more profound understanding of these obstacles, the objective of this paper is to delve into the perspectives and experiences tied to the utilisation of trajectory datasets through a focus group study. This study involves a diverse array of participants from both academia and industry, belonging from various countries and backgrounds. By involving this diverse group, our aim is to uncover the challenges linked to the effective utilisation of trajectory datasets.

A focus group study entails concentrating on specific issues through the participation of a group of individuals engaging in interactive discussions (M. M. Hennink 2013). These discussions fall within the realm of qualitative research tradition, characterised by interactive dialogues among predetermined participants (M. M. Hennink 2013). The primary goal of a focus group discussion is to gather a diverse array of viewpoints on a research topic within a defined timeframe and to establish an atmosphere where participants are at ease expressing their opinions (M. Hennink, Hutter, and Bailey 2011). The focus group methodology distinguishes itself from other qualitative approaches in terms of its objectives, structure, and the collective nature of data collection. Typically consisting of 6 to 8 participants, a focus group study selects individuals based on similar backgrounds or experiences (M. Hennink, Hutter, and Bailey 2011). Furthermore, a designated moderator guides the focus group study, posing questions to stimulate discussions while fostering an open and accepting group environment (M. M. Hennink 2013). The aim is not to reach consensus on the issue but rather to unveil a spectrum of distinct experiences and perspectives.

Through the execution of a focus group study, this qualitative methodology has provided a detailed understanding of the challenges associated with the application of spatio-temporal trajectory datasets. The open-ended questions and concerns raised during the focus group study were initially discussed in our conference paper (Arslan and Cruz 2023). However, it is crucial to connect these challenges with existing literature and propose potential solutions considering state-of-the-art studies and feedback collected during the conference. By doing so, the comprehensive perspective presented in this article will guide the effective utilisation of trajectory datasets across diverse domains. Furthermore, this study enriches the existing body of knowledge on trajectory dataset usage, establishing a foundational foundation for future research on effective

and ethical data utilisation. Ultimately, this effort aims to align the use of trajectory data with ethical and privacy considerations, thereby contributing to the betterment of society.

The paper's organisation is as follows: Section 2 delves into the study's context. Section 3 provides insights into the focus group study, encompassing its setup and participants. Section 4 engages in a conversation about the challenges unearthed through trajectory data usage. Finally, Section 5 wraps up the paper by summarising the findings.

2. Background

This section commences with a definition of the trajectory dataset, providing clarity on its scope and components. Subsequently, it offers an insight into the methodologies employed for generating trajectory datasets, emphasising key approaches and techniques. Following this, it delves into the details of trajectory dataset processing, explaining the methods and tools utilised to refine and prepare the data for analysis. Moving forward, it explores trajectory dataset mining, shedding light on the methodologies employed to extract valuable insights and patterns from the dataset. Finally, it ends by examining the diverse applications of trajectory datasets, showcasing their relevance and impact across various domains.

2.1. Defining trajectory datasets

A spatio-temporal trajectory dataset is a type of data that captures the movement of objects. It is composed of one or more trajectories, where a trajectory represents the trace generated by a moving object in a geographical space, consisting of a series of chronologically ordered points (Yan 2011; Zheng 2015). For example, a trajectory can be represented as $p_1 \rightarrow p_2 \rightarrow p_3 \rightarrow \ldots \rightarrow p_n$, where each trajectory point contains a location indicated by geographical coordinates (x, y) and a corresponding timestamp, such as $p = (x, y, t)$. A typical trajectory dataset comprises five primary fields of trajectory data (see Figure 1). These include the userID, which serves as an identifier for a specific user, the date and time indicating when each trajectory point is collected, the longitude and latitude values representing the geographical coordinates, and contextual information enriching the trajectory point's information (Arslan, Cruz, and Ginhac 2019d). Additionally, trajectory data may include details about the speed and direction of an object's movement (Arslan, Cruz, and Ginhac 2019c). The tracked objects can encompass individuals, vehicles, animals, or any other moving entity.

1	UserID	Datetime	Building Floor	Longitude	Latitude
2	User I	2018-04-19T12:34:17.488	2	5.06808258	47.31020622
3	User I	2018-04-19T12:34:22.591	2	5.06843081	47.31075351
4	User I	2018-04-19T12:34:27.664	2	5.06852338	47.3108042
5	User I	2018-04-19T12:34:32.810	2	5.06845561	47.31079421
6	User I	2018-04-19T12:34:37.866	2	5.06850751	47.31086219

Figure 1. Example of trajectory data.

2.2. Generating trajectory datasets

Trajectory datasets encompass a diverse range of subjects, including people, vehicles, animals, and natural phenomena (net4things 2023; Parent et al. 2013; Ruktanonchai et al. 2018; Shepherd et al. 2021; Unacast 2023; Yan 2011; Zheng 2015) (see Table 1). The generation of trajectory data involves two primary approaches: continuous and episodic recording. Continuous recording facilitates a seamless flow of position information, providing insights into intricate movement patterns, routes, and timing through interpolation. For instance, individuals utilise GPS technology to log their travel routes for personal recollection or to share with friends, while sports enthusiasts, such as cyclists and runners, track their paths to analyse performance. Surveillance cameras are also instrumental in monitoring people's movements and gathering location data. Moreover, connected vehicles, equipped with Light Detection and Ranging

Table 1. Sources of trajectory data (net4things 2023; Parent et al. 2013; Ruktanonchai et al. 2018; Shepherd et al. 2021; Unacast 2023; Yan 2011; Zheng 2015).

No.	Trajectory data type	Description
1	Trajectory of people	People record their movements in the form of trajectories. Examples are; 1) Logging travel routes via GPS to remember journeys and share them with friends. 2) Cyclists and runners documenting their paths for sports analysis. 3) The use of geotagged photos on platforms like Flickr, which can collectively form a trajectory by combining location tags and time stamps. 4) Sequential arrangement of "check-ins" from individuals on location-based social networks, forming a trajectory. 5) Mobile positioning data from CDRs for users carrying their phones. 6) Timestamp and merchant ID data from credit card transactions containing location information. 7) GLH data capturing trajectory details of both local and international trips. 8) Facebook leveraging its location history to track mobility data of active users. 9) Movement monitoring through surveillance cameras, yielding location data.
2	Trajectory of vehicles	1) Connected cars have multiple data sources, including lidar units, radar sensors, and cameras that record location data. 2) GPS-equipped taxis, buses and aircraft record their movements with time information. 3) Trajectory datasets of vehicles are generated by simulating different scenarios for testing and validation of models.
3	Trajectory of animals	Researchers in life sciences use accelerometers, gyroscopes, magnetometers, sensing technologies, etc. for collecting trajectories of animals such as tigers, birds, etc. for studying animals' migratory behaviors and survival situations.
4	Trajectory of natural phenomena	Environmentalists gather trajectories of natural occurrences like tornadoes, hurricanes, and more, aiming to study and forecast natural disasters.

(Lidar) units, radar sensors, and cameras, record their movements for various purposes. Beyond human-related trajectories, researchers in life sciences utilise sensing technologies like accelerometers and magnetometers to collect animal trajectories, enabling the study of migratory behaviours and survival patterns. Furthermore, environmentalists actively gather trajectory data on natural phenomena such as tornadoes and hurricanes to analyse and predict natural disasters.

Conversely, episodic recording methods capture sporadic location snapshots, presenting challenges in restoring complete routes and timing between these discrete points. The limitations inherent in episodic data underscore the importance of carefully weighing the trade-offs between data granularity and the ability to reconstruct comprehensive trajectories. For instance, geotagged photos on platforms like Flickr serve as examples of episodic trajectory data collection, where each photo contains location and time information. Similarly, social media platforms contribute to trajectory data through sequential check-ins on location-based networks, enabling the construction of trajectories over time. Moreover, data from Call Detail Records (CDRs) and credit card transactions, which include timestamp and location information, offer valuable insights into trajectory patterns. Services like Google Location History (GLH) and Facebook location tracking further enrich trajectory datasets by providing detailed information about individuals' local and international trips.

Various sources enable the collection of trajectory data points, as outlined previously. These data points can be plotted on a map, and by connecting them, lines or paths can be formed (see Figure 2). This visualisation technique offers a spatial-temporal representation of an object's movement over time.

2.3. Processing trajectory datasets

Once the trajectory data from diverse sources is stored, it undergoes preprocessing and enrichment processes to prepare it for analysis (Zheng 2015). Preprocessing tasks, including data cleaning, stay point detection, trajectory compression, map matching, and segmentation, are crucial for refining the data quality and structure (Müller and Freytag 2005). Data cleaning filters noise, stay point detection identifies significant locations, compression reduces data volume, map matching aligns trajectories with road networks, and segmentation breaks trajectories into meaningful segments. These processed trajectories are then enriched with contextual information to create semantic trajectories, enhancing their analytical value. Several software packages (Refer to Table 2, adapted from the research conducted by Martin et al. (2023)) have been developed to meet the diverse requirements of processing and analysing trajectory data.

Prominent among them are Scikit-mobility (Pappalardo et al. 2019), PyMove (Sanches 2019), MovingPandas (Graser 2019), PTRAIL (Haidri et al. 2022), Movetk

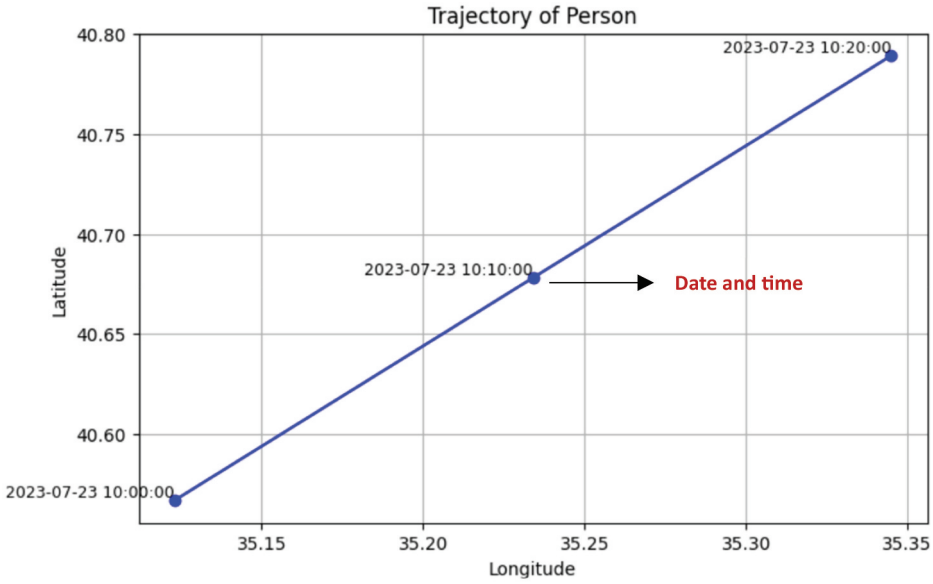

Figure 2. Plotting trajectory data.

(MoveTK 2023), TraClets (Kontopoulos, Makris, and Tserpes 2023), Trackintel (Martin et al. 2023), Mat-Builder (Pugliese et al. 2022), Traja (Shenk et al. 2021), Traffic (Olive 2019), MovinPy (Theodoropoulos and Theodoridis 2018), HuMobi (Smolak, Rohm, and Sila-Nowicka 2022), TransBigData (Q. Yu and Yuan 2022), Trajectory (Custódio, Jansen, and Chaitanya 2020), Yupi (Reyes et al. 2023), Automatize (Portela et al. 2022), Pymeos (Pymeos PyPI 2024), and Stplanr (Lovelace and Ellison 2018). These libraries offer a range of functionalities including data preprocessing, noise filtering, stop detection, trajectory compression, segmentation, outlier detection, clustering, visualisation, and semantic enrichment (Martin et al. 2023).

2.4. Mining trajectory datasets

Once trajectory datasets are processed, they unlock numerous trajectory data mining tasks aimed at discovering insightful knowledge, such as patterns, from the data (Mazimpaka and Timpf 2016). These tasks encompass trajectory data clustering, classification, pattern mining, outlier detection, and prediction (Yan 2011; Zheng 2015). Trajectory data clustering categorises trajectories into clusters based on similar movement characteristics, providing insights into groupings within the dataset. Classification involves assigning objects into predefined classes, enabling the identification of patterns based on labelled samples. Trajectory pattern mining uncovers hidden movement patterns, categorising them into repetitive, frequent, or group patterns to understand when, where, and which entities are involved. Outlier detection identifies trajectories

Table 2. Trajectory dataset processing software packages with functionalities.

			Purpose					
No.	Package name	Programming language	Preprocessing	Classification/Clustering	Compression	Visualization	Privacy risk assessment	Semantic enrichment
1	Scikit-mobility	Python	✓	X	✓	✓	X	X
2	PyMove	Python	✓	✓	✓	X	X	X
3	MovingPandas	Python	✓	X	X	✓	X	X
4	PTRAIL	Python	✓	✓	X	✓	X	X
5	Movetk	C++	✓	✓	X	✓	X	X
6	TraClets	Python	X	✓	X	X	X	X
7	Trackintel	Python	✓	X	X	✓	X	✓
8	Mat-Builder	Python	✓	X	✓	X	X	✓
9	Traja	Python	✓	✓	X	✓	X	X
10	Traffic	Python	✓	✓	X	✓	X	X
11	MovinPy	Python	✓	X	✓	X	X	✓
12	HuMobi	Python	✓	✓	✓	X	X	X
13	TransBigData	Python	✓	✓	X	✓	X	X
14	Trajectory	Python	X	X	✓	X	X	X
15	Yupi	Python	✓	X	X	✓	X	X
16	Automatize	Python	✓	✓	X	✓	X	X
17	Pymeos	Python	✓	X	X	X	X	X
18	Stplanr	R	✓	X	X	✓	X	X

(✓ Available, X Not available).

that deviate from the dataset's general behaviour, focusing on rare patterns rather than common ones found in pattern mining. Finally, prediction utilises trajectory data to forecast the future locations of moving objects, offering valuable insights for various applications.

2.5. Applications of trajectory datasets

Trajectory data mining plays a crucial role in tackling various application challenges related to trajectories (Mazimpaka and Timpf 2016). In the realm of social dynamics, understanding social relationships, such as ties between individuals and communities, involves clustering individuals' stay locations and detecting social events through spatiotemporal clustering. Additionally, characterising connections between places relies on describing links, with trajectory outlier identification playing a role in discerning abnormal links. In transportation, identifying origins, destinations, and movement patterns depends on stay location identification and trajectory pattern mining techniques. Similarly, urban planning requires understanding mobility characteristics to facilitate efficient city design, employing trajectory classification. Environmental applications quantify pollution levels by utilising trajectory pattern mining for effective control measures, while energy sector analysis utilises similar methods to study consumption patterns. It is important to note that there is not always a one-to-one mapping between an application issue and mining method; multiple methods may be needed for a single issue, or different issues may be tackled using the same method. Apart from data mining-based applications, trajectory datasets are also explored for providing visual analytics. Several systems have been developed to address various analytical needs, including RoseTrajVis for studying commuting behaviours (Ferreira et al. 2022) and (Chae et al. 2015)'s system for analysing anomalous movements during disasters. VISTA (Soares et al. 2019) supports trajectory data exploration and processing, SmartAdP (Liu et al. 2016) integrates visualisation-driven data mining for tasks like retail store location selection, and UrbanMotion (Shi et al. 2020) offers insights into urban mobility patterns. These diverse applications underscore the versatility and significance of trajectory datasets across domains (D. Wang, Miwa, and Morikawa 2020).

3. Focus group study

Undoubtedly, trajectory data holds immense potential for various applications. However, the effective utilisation of this data comes with its own set of challenges. To delve into these challenges, a focus group methodology was employed. The focus group consisted of 49 participants in the field of Computer and Information Science, hailing from diverse countries (as depicted in Figure 3) including Denmark (DK), Netherlands (NL), Italy (IT), Germany (DE),

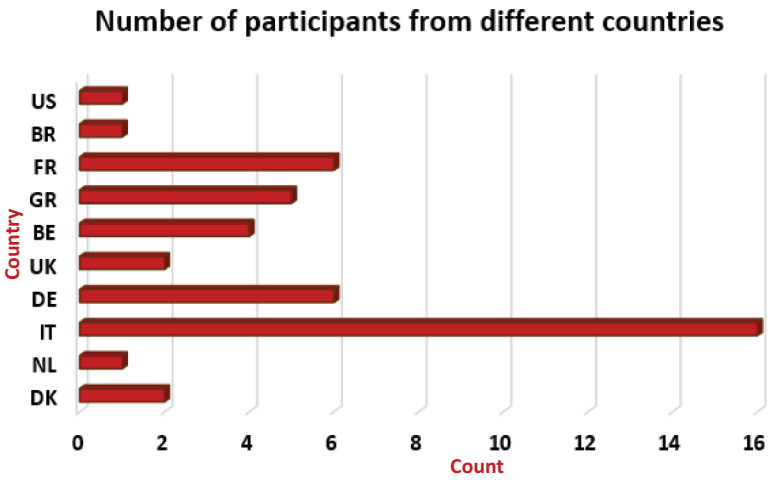

Figure 3. Diversity of participants.

United Kingdom (UK), Greece (GR), France (FR), Brazil (BR), and the United States (US). The participants encompassed Ph.D. students, researchers, and practitioners working in mobility data science. The focus group was moderated by Konstantinos Tserpes from Harokopio University, Greece, and featured panellists such as Thierry Chevallier (AKKA Technologies, France), Bettina Berendt (Technische Universität Berlin, Germany & KU Leuven, Belgium), Franco Maria Nardini (ISTI-CNR, Italy), Luca Pappalardo (ISTI-CNR, Italy), and Mahmoud Sakr (Université Libre de Bruxelles, Belgium).

The primary objective of the focus group was to comprehensively explore the various challenges associated with the use of spatio-temporal trajectory datasets. This two-hour discussion took place as part of the 1st International Summer School on Data Science for Mobility, held in Santorini, Greece, from 3 to 7 October 2022. The summer school was organised under the MASTER (Multiple ASpects TrajEctoRy management and analysis) project, funded under the H2020-MSCA-RISE-2017 call. This project aimed to establish an international and inter-sectoral network of organisations engaged in a joint research programme to develop new methods for building, managing, and analysing semantic trajectories.

4. Trajectory dataset challenges

During the focus group discussions, a comprehensive set of challenges emerged (see Table 3), encompassing various issues and open research questions surrounding spatio-temporal trajectory datasets (Arslan and Cruz 2023). These challenges reflect the complexities inherent in utilising such datasets effectively. One of the key challenges discussed was understanding the factors that motivate mobile users to willingly share their data, considering the value proposition

Table 3. Trajectory dataset challenges.

No.	Challenges (issues/open research questions)	Type	Concerning
1	How can we convince mobile users to share their data for a value against a contract?	Open question	Dataset construction
2	Do people come under pressure when their trajectory data is monitored?	Open question	
3	Can the idea of a traffic simulation or agent-based model be used to generate trajectory datasets?	Open question	
4	Is constructing a synthetic trajectory dataset preferable over a real trajectory dataset?	Open question	
5	There is no benchmark that exists to develop and publish trajectory datasets.	Concern	
6	How can we make trajectory datasets findable, accessible, interoperable, and reusable?	Open question	
7	There is no standard that exists to preprocess the trajectory datasets.	Concern	Dataset preprocessing
8	There is no standard that exists to extract spatio-temporal and kinematic features from trajectory datasets.	Concern	
9	Can we protect data privacy by sharing the inferred knowledge than real trajectory datasets?	Open question	Dataset privacy
10	Can we use the help of social scientists to define the norms for respecting the trajectory data privacy and ethics?	Open question	
11	Can we share trajectory datasets directly with researchers rather than with companies?	Open question	Dataset sharing
12	Can downloading the trajectory dataset from public machines help to avoid data leaks?	Open question	
13	Can synthetic datasets overfit the trajectory prediction models for future analysis?	Open question	Dataset use
14	There is no policy that exists to use a private trajectory dataset in research.	Concern	

and contractual agreements involved. Additionally, the potential psychological pressure experienced by individuals when their trajectory data is under monitoring raised concerns about privacy and personal autonomy. Exploring the concept of traffic simulation or agent-based models as a means of generating trajectory datasets sparked discussions on the technical and practical aspects of implementation. Another critical question that arose was whether synthetic trajectory datasets are a preferable alternative to real datasets. This challenge considers the trade-offs between realism and privacy, as well as the applicability of synthetic data in research and practical applications. The absence of a benchmark for developing and publishing trajectory datasets was recognised as an obstacle, making it challenging to evaluate and compare different approaches. Moreover, ensuring the findability, accessibility, interoperability, and reusability of trajectory datasets emerged as an essential challenge in enabling their effective utilisation. Standardisation was identified as a pressing need, particularly in terms of preprocessing trajectory datasets and extracting meaningful spatio-temporal and kinematic features.

The group also explored novel approaches to data privacy, discussing the potential of sharing inferred knowledge instead of real trajectory datasets. This approach not only protects individual privacy but also involves interdisciplinary collaboration with social scientists to define ethical norms and guidelines. The question of sharing trajectory datasets directly with researchers, rather than relying on intermediaries such as companies, was raised to

foster a more open and collaborative research environment. Additionally, exploring the benefits of downloading trajectory datasets from public machines to mitigate data leaks and privacy concerns was highlighted. The potential for synthetic datasets to overfit trajectory prediction models prompted discussions on the appropriate usage and validation of such datasets. Lastly, the lack of policies governing the use of private trajectory datasets in research was identified as an important issue, necessitating ethical guidelines and regulations.

The following sections provide a detailed explanation of each challenge, delving into the intricacies and considerations associated with addressing these issues.

4.1. Trajectory dataset construction

There are various sources available for generating trajectory data (as mentioned in Table 1), each offering unique advantages and capturing different aspects of movement patterns (Yan 2011; Zheng 2015). Researchers can leverage these sources independently or in combination to generate comprehensive spatio-temporal trajectory datasets tailored to their specific research questions. The selection of data sources depends on several factors, such as the nature of the research question and the availability of data. Researchers must carefully consider which sources are most relevant and provide the necessary granularity and coverage to address their specific objectives. For instance, GPS devices are ideal for tracking individual movement patterns, while social media platforms can offer insights into collective mobility behaviour. Additionally, combining multiple data sources can provide a more holistic understanding of trajectory patterns and enhance the richness of the generated datasets. However, constructing trajectory datasets is not without challenges. Through a comprehensive focus group study, several common issues have been identified, including:

4.1.1. Challenge 1: How can we convince mobile users to share their data for a valueagainst a contract?

The primary obstacle emerging in relation to constructing trajectory datasets lies in how to persuade mobile users to share their data for research purposes. Encouraging mobile users in motion to share their trajectory data in exchange for value within a contractual framework presents a formidable challenge. Nonetheless, the existing literature (Bu-Pasha et al. 2016; Hox and Boeije 2005) reveals various strategies that can increase the likelihood of success. A few of these strategies are outlined below:

(1) Transparency and informed consent: Researchers must adopt an approach of openness and clarity concerning data collection practices

to safeguard user privacy. This entails acquiring informed consent from users. This process involves providing users with clear details about data collection, its purpose, and allowing them the option to participate or decline.

(2) Value proposition: To encourage users to share their trajectory data, tangible benefits must be extended. These benefits might encompass discounts, personalised recommendations, additional services, or rewards.

(3) Trust and security: Establishing user trust is crucial, which can be achieved by assuring them that their data will be managed securely. It is important to convey that privacy and security regulations, such as the European Union's General Data Protection Regulation (GDPR), will be strictly followed during the data collection process (Bu-Pasha et al. 2016).

4.1.2. Challenge 2: Do people come under pressure when their trajectory data ismonitored?

Despite implementing various strategies to persuade users to share their trajectory data and foster trust, the process of data collection can still impose a burden on them. Users might experience discomfort, anxiety, or a sense of insecurity due to the perception of being under constant surveillance. This unease can be particularly pronounced when the collected data is utilised for impactful decision-making, such as targeted advertising, risk assessment, or law enforcement activities (Phillips 2018).

To alleviate this burden, users require reassurance from researchers that their data will undergo anonymisation to safeguard their privacy. Additionally, it is imperative for research institutions and universities to establish well-defined data governance policies (Nill and Aalberts 2014), encompassing aspects like data security, sharing protocols, and data retention practices. These measures collectively contribute to ensuring the ethical and legal utilisation of the data, thereby mitigating the pressures users may experience.

4.1.3. Challenge 3: Can the idea of a traffic simulation or agent-based model beused to generate trajectory datasets?

In the context of avoiding the intricate processes of convincing individuals, establishing their trust, and ensuring legal use of their data, the concept of a traffic simulation or agent-based model provides a solution for generating synthetic trajectory datasets. A traffic simulation or agent-based model is a computational approach used to simulate the behaviour and interactions of entities, such as vehicles or pedestrians, within a given environment (Balmer et al. 2004). These models replicate real-world scenarios and dynamics to generate realistic movement patterns and trajectories (Balmer et al. 2004). By utilising traffic simulation or agent-based models, it becomes possible to generate synthetic spatio-temporal trajectory datasets. These datasets mimic the

trajectories and patterns that would be observed in real-world scenarios. They provide valuable insights and information without relying on actual data collected from individuals or objects. This approach of using synthetic trajectory datasets becomes particularly advantageous when obtaining real-world data is challenging or unavailable. In situations where gathering authentic data is difficult, costly, or raises privacy concerns, traffic simulation or agent-based models offer a viable alternative. They allow for the generation of realistic synthetic data that can be used for various purposes such as research, analysis, testing, or training models without compromising the privacy or security of individuals or entities.

4.1.4. Challenge 4: Is constructing a synthetic trajectory dataset preferable overa real trajectory dataset?

The synthetic data generated by traffic simulation or agent-based models serves multiple purposes, including testing and validating trajectory analysis algorithms and training machine learning models. These applications benefit from having access to realistic yet privacy-preserving datasets. Trajectory analysis algorithms, which aim to extract insights and patterns from movement data, can be evaluated using synthetic trajectory datasets created by traffic simulation or agent-based models. These datasets provide a controlled environment for algorithm testing, allowing researchers to assess the performance and accuracy of their algorithms without the need for real-world data. This enables algorithm refinement and optimisation before applying them to real-world scenarios. Furthermore, machine learning models can be trained using synthetic trajectory datasets from traffic simulation or agent-based models. These datasets provide a source of labelled data that can be used to train models to recognise and predict various trajectory patterns. By training on synthetic data, models can learn to make accurate predictions and classifications without requiring access to sensitive or proprietary real-world data.

However, it is important to note that synthetic datasets generated by traffic simulation or agent-based models may not always perfectly represent real-world conditions. The virtual environment of a traffic simulation or agent-based model, while designed to mimic reality, may have limitations that affect the fidelity of the generated data. These limitations can arise from factors such as simplified modelling assumptions, incomplete understanding of real-world dynamics, or uncertainties in the simulation process. Therefore, caution should be exercised when extrapolating insights or conclusions solely based on synthetic datasets. To address these limitations, it is crucial to validate synthetic data against real-world data whenever possible. By comparing the synthetic trajectories with actual trajectories collected from real individuals or objects, researchers can assess the accuracy and reliability of the traffic simulation or agent-based model. This validation process helps identify any discrepancies or biases in the synthetic data and provides a benchmark for evaluating its quality.

4.1.5. Challenge 5: There is no benchmark that exists to develop and publish 415trajectory datasets.

Whether derived from the physical world or generated through traffic simulation or agent-based models, the development and release of trajectory datasets necessitate adherence to established benchmarks. According to focus group participants, the absence of a universally recognised benchmark for creating trajectory datasets is obvious. This absence could stem from various factors, such as: 1) Divergent data formats: The lack of a standardised format for storing trajectory data results from different research groups adopting distinct data structures. 2) Complex and varied nature: The multifaceted composition of trajectory datasets, gathered from diverse sources and encompassing varied information types, poses challenges in standardising datasets for widespread applications. 3) Proprietary constraints: Many trajectory datasets remain proprietary, accessible exclusively to their creators. This proprietary nature restricts their utility as benchmarks.

Nevertheless, certain trajectory datasets within the research community can serve as benchmarks. Notable examples encompass the Geolife GPS trajectory dataset (Zheng, Xie, and Ma 2010), Stanford Drone dataset (Andle et al. 2022), ETH Zurich datasets (Zurich and Datasets 2023), UCY dataset (Lerner et al. 2007), and an indoor crowd movement trajectory benchmark dataset introduced in (Zhao et al. 2021). Typically combined from drones or mobile platforms, these datasets incorporate attributes like GPS coordinates, timestamps, and sensor readings.

4.1.6. Challenge 6: How can we make trajectory datasets findable, accessible, inter-operable, and reusable?

Upon the development of trajectory data, a crucial step is to render it discoverable, attainable, interoperable, and open to reuse (Herman and Heidmann 2002; Wilkinson et al. 2016). This procedure is pivotal to ensuring the data's utility across a spectrum of research applications. To render datasets discoverable, their presence should be catalogued within pertinent data repositories such as the Global Earth Observation System of Systems (GEOSS) (Nativi et al. 2015) or the European INSPIRE Geoportal (Crompvoets et al. 2004). These listings should include comprehensive descriptions through standard metadata fields like keywords, abstracts, and contact details.

To ensure broad accessibility, trajectory datasets should embrace open formats such as Comma-Separated Values (CSV) or JavaScript Object Notation (JSON). These formats should be made available through user-friendly interfaces like RESTful Application Programming Interfaces (APIs) or web portals (Ehsan et al. 2022). The goal is to make the data accessible not only to researchers and developers but also to the public. Achieving interoperability is equally crucial, requiring the formatting of trajectory datasets to align seamlessly with other datasets and systems. This

alignment can be achieved by adhering to universally recognised data standards, such as the Open Geospatial Consortium (OGC) standards (van Rees 2013) for geospatial data or the International Standard Organization (ISO) 19115 standard for metadata (Růžička 2008).

The ISO proposes 'ISO19141:2008 geographic information – Schema for moving features' as an abstract data model for describing trajectory data (International Organization for Standardization 2008). However, this model lacks specific implementation schemas like Extensible Markup Language (XML) or JSON. Recognising the practical need for an implementation standard that facilitates actual data exchange, a new standard called 'OGCR Moving Features' was introduced. This standard was subsequently adopted by the OGC in February 2015, aiming to provide a practical solution for encoding trajectory data and overcoming the limitations of the abstract ISO19141:2008 standard (Asahara et al. 2015).

Facilitating dataset reuse requires comprehensive documentation, including data collection methodologies, data quality insights, and any inherent limitations. This documentation should be integrated into standardised metadata formats. It is essential to underscore that achieving discoverability, accessibility, interoperability, and reusability for spatio-temporal trajectory datasets encompasses not only technical aspects but also encompasses legal, ethical, and organisational considerations. Thus, establishing clear dataset-sharing agreements, data governance policies, and protocols is crucial, along with ensuring alignment with data privacy, security, and intellectual property laws and regulations.

4.2. Trajectory dataset preprocessing

Both synthetic and real-world datasets must undergo preprocessing prior to their utilisation in research (Haidri et al. 2022; D. Wang, Miwa, and Morikawa 2020). Preprocessing serves as a crucial stage in rendering data amenable for analysis, ultimately enhancing result accuracy, efficiency, and interpretability (Haidri et al. 2022). However, trajectory data preprocessing can be a laborious and time-intensive endeavour, contingent upon the specific application use case (Haidri et al. 2022). Preprocessing trajectory datasets involves several critical stages to ensure data quality and usability. These stages, outlined in established practices, include data cleansing to address anomalies and ensure real-world accuracy, and data transformation to format trajectory data for analysis. Additional preprocessing steps encompass stay point detection, trajectory interpolation, map matching, trajectory segmentation, trajectory compression, and data anonymisation. Each step plays a vital role in enhancing data quality, managing storage overhead, and safeguarding individual privacy.

4.2.1. Challenge 7: There is no standard that exists to preprocess the trajectory datasets.

The focus group highlighted a significant concern regarding the preprocessing of trajectory datasets: the lack of a universally accepted method within existing literature. Methodologies and techniques described in the literature vary depending on factors such as data type, research queries, and intended data applications. As a result, the outcomes of these diverse approaches often differ across studies due to the heterogeneous nature of datasets and the utilisation of distinct definitions grounded in individual interpretations. This variation has led to the development of an extensive array of application-specific preprocessing methods for trajectory datasets. Moreover, even among published trajectory datasets, specific details regarding preprocessing steps and techniques are often missing, leaving researchers unaware of the methods used to generate the datasets.

Numerous software libraries have emerged to address the diverse requirements of processing and analysing trajectory data, including Scikit-mobility (Pappalardo et al. 2019), PyMove (Sanches 2019), MovingPandas (Graser 2019), PTRAIL (Haidri et al. 2022), Movetk (MoveTK 2023), TraClets (Kontopoulos, Makris, and Tserpes 2023), Trackintel (Martin et al. 2023), and Mat-Builder (Pugliese et al. 2022). However, it is essential to note that these libraries do not adhere to a standardised approach for executing data processing tasks. Some libraries employ various filters to clean noisy data, while others may not consider filtering at all. Similarly, while some libraries output stay points, others do not include this feature. The lack of a universally agreed-upon series of preprocessing tasks for trajectory data analysis is evident. Furthermore, existing studies in literature also demonstrate a lack of standardised rules for trajectory data preprocessing. For instance, Feng et al. (2018)'s DeepMove model uses deep learning for next-location prediction without preprocessing raw position records, whereas Urner et al. (2018) and Arslan et al. (2019d) extract stay points from trajectory datasets. These discrepancies underscore how the absence of standardised trajectory model definitions and common preprocessing practices hampers the reproducibility and comparability of techniques and analysis outcomes. To address this issue, the establishment of a standardised approach is crucial in the development of trajectory datasets.

4.2.2. Challenge 8: There is no standard that exists to extract spatio-temporal and kinematic features from trajectory datasets.

During the preprocessing phase of trajectory datasets, the extraction of spatio-temporal and kinematic attributes from trajectory datasets has been observed in existing research as well (Arslan et al. 2018; Arslan, Cruz, and Ginhac 2019b; Flash, Meirovitch, and Barliya 2013; Omlor and Giese 2007). Temporal features entail characteristics drawn from the

temporal dimension, such as month, week, day, hour, etc. Semantic features encompass contextual information that aligns with movement based on environmental factors. An illustration of a semantic feature is the identification of a point of interest visited within a building or city. Lastly, kinematic features (D. Wang, Miwa, and Morikawa 2020) pertain to attributes concerning the object's motion, like speed, rotation, etc. The focus group has substantiated that a standardised methodology for extracting spatio-temporal and kinematic features from trajectory datasets is currently lacking. This absence of a standard arises from the fact that the feature extraction process hinges on the distinct research application and data type. Given the diversity of applications and data types, certain features might hold greater relevance or utility than others. Hence, the selection of features must align closely with the specific task at hand.

4.3. Trajectory dataset privacy

Trajectory datasets have the potential to raise privacy concerns due to their inclusion of an individual's time-varying location information (Nergiz, Atzori, and Saygin 2008). This information can be harnessed to deduce sensitive insights into an individual's personal routines, habits, and relationships (Zheng, Xie, and Ma 2010). For instance, it can lead to the disclosure of an individual's home or workplace, real-time tracking of their movements, deductions about their behaviours, health status, or social associations (Basu et al. 2014). Furthermore, even when trajectory data is anonymised, it might still harbour adequate information to facilitate re-identification of individuals, particularly when merged with other data sources (Monreale et al. 2014). Beyond these concerns, trajectory data possesses the potential for enabling surveillance over individuals or groups, potentially imposing civil authorities and human rights. The subsequent section outlines the challenges intrinsically linked to privacy matters in trajectory datasets.

4.3.1. Challenge 9: Can we protect data privacy by sharing the inferred knowledge than real trajectory datasets?

To address privacy-related concerns, the focus group has proposed an approach of sharing inferred insights derived from spatio-temporal trajectory datasets, as opposed to sharing the raw data itself. Inferences drawn from the data enable the removal of sensitive details like individual identities and specific locations. However, it is essential to acknowledge that the level of privacy safeguarding offered by inferred knowledge hinges on the techniques employed and the extent of detail contained in the inferred knowledge. Nevertheless, it is noteworthy that particular forms of inferred knowledge, such as aggregate statistics or patterns, might still unveil sensitive particulars about individuals or groups

(Salah, Canca, and Erman 2022). Additionally, relying solely on sharing inferred knowledge may not universally suffice in ensuring data privacy. Instances exist where combining such inferred knowledge with other information sources could potentially lead to the re-identification of individuals or groups within the dataset (Salah, Canca, and Erman 2022). Moreover, if the techniques used for inferring knowledge are not resilient against potential attacks, the original data might be reverse engineered by malicious entities.

4.3.2. Challenge 10: Can we use the help of social scientists to define the norms forrespecting trajectory data privacy and ethics?

In addition to data privacy and security experts, the involvement of social scientists is essential in tackling ethical and privacy challenges associated with trajectory datasets (Salah, Canca, and Erman 2022). Their valuable contributions extend beyond technical expertise and involve a deeper understanding of the societal implications. Social scientists can aid in identifying potential ethical dilemmas that may arise from trajectory data collection and propose effective solutions to mitigate them. Moreover, they possess the ability to comprehend the social and cultural factors that shape individuals' attitudes and beliefs towards data gathering and utilisation. Furthermore, social scientists can contribute to the establishment of equitable and responsible norms and protocols for data governance within organisations. By considering the perspectives of all stakeholders involved, they ensure that data practices align with fairness, transparency, and accountability. This interdisciplinary collaboration enables a comprehensive and holistic approach to address privacy concerns, promoting ethical practices and safeguarding individual rights in the preprocessing of trajectory datasets.

4.4. Trajectory dataset sharing

Trajectory dataset sharing involves the act of making trajectory datasets accessible to the public (Zheng 2015). This practice allows researchers and developers to utilise these datasets, extract valuable insights, and create innovative applications for further research. Additionally, sharing trajectory datasets fosters openness and transparency in the scientific community, enabling others to validate and verify research findings. Through extensive focus group discussions, the following challenges have been identified in relation to trajectory dataset sharing:

4.4.1. Challenge 11: Can we share trajectory datasets directly with researchers 615 rather than with companies?

The focus group raised a pivotal query: Is it prudent to directly share trajectory datasets with researchers as opposed to companies? This practice can serve to avoid commercial gain. Researchers often pursue specific hypotheses or testing

objectives and may not be driven by the same commercial motives as companies. Distributing datasets directly to researchers can boost the assurance that data deployment aligns with scientific and academic pursuits rather than commercial interests. Furthermore, it is worth emphasising that sharing spatio-temporal trajectory data directly with researchers can yield advantages for the owning company or organisation. Researchers frequently generate insights and novel knowledge that can enhance products, services, and policies, thereby bestowing benefits upon the entity. In the process of data sharing, meticulous attention to data privacy and security risks is paramount. To preserve individual privacy, the dataset should undergo de-identification. Moreover, secure channels like fortified data repositories or virtual private networks should facilitate data exchange, preventing unauthorised access. However, establishing a reasoned data sharing agreement remains essential. This document should define the terms and conditions governing data sharing, including research objectives, data utilisation, privacy safeguards, and the mechanism for disseminating research outcomes.

4.4.2. Challenge 12: Can downloading the trajectory dataset from public machineshelp to avoid data leaks?

As suggested by the focus group, one approach to mitigate the risks of data leaks is to obtain spatio-temporal trajectory datasets from publicly accessible machines within universities or research institutes. By 'publicly accessible machines', we refer to managed computers within these institutions that adhere to institutional Information Technology (IT) security policies, in contrast to private machines like researchers' personal laptops that are not centrally managed. The rationale behind this recommendation is rooted in the belief that personal machines, such as laptops, are more susceptible to theft or loss, potentially exposing sensitive data. Utilising publicly accessible machines within secured institutional environments may provide a level of centralised control and monitoring.

However, it is crucial to acknowledge that downloading datasets from public machines does not completely eliminate the possibility of data breaches (Tu et al. 2018). The effectiveness of this approach heavily relies on the security measures implemented on these public machines. In cases where public machines lack robust data security protocols, the risks associated with data leaks may be worsened rather than mitigated. To enhance the security level of public machines used for dataset downloads, it is imperative to implement advanced security measures. This includes the deployment of state-of-the-art firewalls to reinforce the network perimeter, access controls to restrict unauthorised access, and intrusion detection systems to promptly identify and respond to any potential security incidents. Additionally, regular security audits should be conducted to assess the

effectiveness of these security measures and identify any vulnerabilities that may need attention.

4.5. Trajectory dataset use

Leveraging trajectory datasets for diverse research applications presents challenges at both the individual trajectory level and the trajectory database level. One prominent challenge at the trajectory level is data sparsity, resulting from gaps within the data that hinder accurate inference of movement patterns. These gaps often stem from missing data due to issues such as battery, storage, or transmission-related problems, as well as low spatial or temporal resolution, which adds further complexity (Z. Wang, Zhang and James 2020). To address this challenge, trajectory reconstruction processes (Li et al. 2019; Marketos et al. 2013; Tong et al. 2021; F. Yu et al. 2022) aim to complete individual movement information by recovering unspecified positions. Several efforts have been made in literature to tackle these issues. For example, a recurrent neural network-based encoder-decoder deep learning approach (D. Wang, Miwa, and Morikawa 2020) has been proposed, alongside offline reconstruction methods for missing vehicle trajectory data from 3D LIDAR (Sazara, Nezafat, and Cetin 2017). Additionally, Kamel, a scalable trajectory imputation system, has been developed to insert additional realistic trajectory points, thereby enhancing the accuracy of trajectory applications (Musleh and Mokbel 2023).

Another challenge is data quality, where noise and errors compromise data reliability (Andrienko, Andrienko, and Fuchs 2016; Graser 2021). Noise filtering algorithms, categorised into Mean or Median filter, Kalman and Particle filter, and Heuristics-Based Outlier Detection, are employed to enhance accuracy (Zheng 2015). Using trajectory datasets also involves challenges in data visualisations due to vast information across dimensions. Traditional methods, like contour and shading maps, focus on spatiotemporal information, while evolving techniques consider additional attributes like speed and orientation. For an in-depth exploration of trajectory visualisation methods, reference to state-of-the-art literature is recommended (Gonçalves, Afonso, and Martins 2013).

At the trajectory database level, several challenges arise;

1) Compression: Storing extensive trajectory data generated by location-acquisition techniques poses challenges due to redundancy. Trajectory compression algorithms show promise in reducing storage requirements and communication loads. However, these algorithms often involve a tradeoff between compression ratio and maximum error. Higher compression ratios may result in poorer quality of compressed trajectory data.

2) Scalability: Managing vast trajectory datasets on a large scale demands significant computational resources and time, leading to computational expense and time-consuming operations. Traditional trajectory management systems, such as PostgreSQL, Oracle, HDFS, and Azure, which are disk-oriented,

can face challenges related to scalability and slow query processing (Ribeiro de Almeida et al. 2020). To address this, the adoption of Big Data platforms like Apache Spark has become increasingly common for the efficient management of trajectory data (Ribeiro de Almeida et al. 2020).

3) Data integration: Integrating data from diverse sources introduces complexities, including variations in data formats, temporal and spatial resolutions, and coordinate systems within trajectory databases (Ribeiro de Almeida et al. 2020). Recent research (Heath and Bizer 2022) focusing on modelling trajectory data using RDF graphs or ontologies has gained attention. This approach proves beneficial for integrating trajectory data with other datasets, enabling semantic enrichment case studies. The representation of semantic trajectory data using RDF not only facilitates the inference of new knowledge but also allows the publication of data as Linked Open Data (LOD), enhancing accessibility on the Semantic Web (Heath and Bizer 2022). 4) Privacy concerns present significant hurdles, especially in human tracking scenarios where high-quality trajectory data availability is constrained due to privacy considerations. While animal trajectories are often openly accessible, safeguarding the privacy of human trajectory data remains paramount for researchers. Tackling these challenges comprehensively is essential for advancing trajectory data applications across diverse domains (Ribeiro de Almeida et al. 2020).

4.5.1. Challenge 13: Can synthetic datasets overfit the trajectory prediction models for future analysis?

The focus group has expressed concerns concerning the utilisation of synthetic trajectory datasets. A pivotal concern, raised based on their prior experiments, is the prevalence of overfitting when employing synthetic trajectory datasets for the training and testing of trajectory prediction models. Overfitting arises when a model becomes exceedingly attuned to the training data, resulting in subpar performance on new and previously unseen data (Schittenkopf, Deco, and Brauer 1997). This can transpire when the synthetic dataset fails to accurately mirror real-world conditions the model will ultimately encounter, or if the model's complexity surpasses available data.

Existing literature (Schittenkopf, Deco, and Brauer 1997) offers various strategies to counter the partiality of synthetic trajectory datasets to overfit prediction models. One approach involves ensuring that synthetic datasets closely emulate the real-world conditions where the model is destined to operate. This alignment safeguards against the model becoming overly reliant on the training data. Additionally, the exploration of cross-validation – a method wherein the dataset is partitioned into multiple subsets, and the model is trained and assessed on each subset – can also be considered to curtail the likelihood of overfitting. Moreover, the application of regularisation techniques proves effective in averting

overfitting. Further insights on regularisation methods can be explored here (Santos and Papa 2022). Ultimately, opting for simpler models with fewer parameters can also mitigate the risk of overfitting, as it reduces the model's susceptibility to becoming too finely tuned to the training data.

4.5.2. Challenge 14: There is no policy that exists to use a private trajectory dataset in research.

To address the challenge of mitigating model overfitting with synthetic datasets, the existing literature also explores the utilisation of private real-world trajectory datasets (Huang et al. 2019). However, the focus group underscores the absence of a universally embraced protocol governing the incorporation of private trajectory datasets into research endeavours. This absence can be attributed to several factors. One primary reason stems from the nature of trajectory datasets, which often encompass sensitive information. Researchers must be well-versed in data usage agreements and legal requisites. In certain instances, obtaining ethical clearance from institutional review boards may be obligatory prior to employing private trajectory datasets. This intricacy complicates the formulation of a standardised policy that effectively balances the imperatives of data privacy and research data access. Moreover, shaping a comprehensive policy for incorporating private trajectory datasets into research endeavours demands substantial resources, including legal proficiency, data privacy acumen, and in-depth comprehension of the research domain. Lastly, the absence of a universal norm for collecting, storing, and preprocessing trajectory data further complicates the establishment of a policy applicable across diverse private datasets. Lastly, the challenge lies in crafting a policy that accommodates the intricacies of data privacy while facilitating data access for research purposes.

5. Discussion and future directions

The rapid advancement in the collection of spatio-temporal trajectory data has brought significant benefits to various industries (Mokbel et al. 2022; Sakr, Ray, and Renso 2022). However, alongside its potential, the utilisation of trajectory datasets also faces numerous challenges, which are comprehensively addressed in this article. This study adopts a unique approach by employing the focus group method, which has not yet been explored in the current state-of-the-art literature, to investigate challenges associated with the utilisation of trajectory datasets. By employing this approach, we have successfully identified and documented a comprehensive set of 14 challenges associated with various aspects of trajectory dataset construction, preprocessing, privacy considerations, sharing, and utilisation. Below, we present our insights and recommendations aimed at addressing these identified challenges.

(1) **Benchmark Development**: One of the significant challenges highlighted pertains to the absence of a benchmark for the development and publication of trajectory datasets. The creation of benchmarks for trajectory datasets stands as a pivotal milestone in advancing research within the realm of mobility data analysis. To craft meaningful benchmarks, it is imperative to initiate the process with the comprehensive collection of diverse trajectory datasets, spanning across a spectrum of scenarios, including vehicular movements, pedestrian pathways, and even animal migrations. These datasets should encompass variations in terms of size, complexity, and geographical locations to faithfully represent real-world scenarios. Furthermore, the development of such benchmarks requires a precise approach that considers not only the trajectories but also the contextual information linked to the environment in which these trajectories occur. This contextual information is crucial for different trajectory scenarios, whether indoors or outdoors.
   (a) For indoor trajectory scenarios, having access to building information is utmost. Numerous building data platforms are available (Arslan, Cruz, and Ginhac 2019b), offering a wealth of building-related information for research and benchmarking purposes. For instance, Building Information Modelling (BIM) (Arslan, Cruz, and Ginhac 2019a) provides a robust platform for acquiring detailed building information, including floor plans, structural layouts, and room configurations. Incorporating BIM data into trajectory datasets enables researchers to analyse and develop mobility solutions tailored to indoor environments such as airports, shopping malls, and office buildings.
   (b) On the other hand, for outdoor trajectory scenarios, gaining a deep understanding of the urban environment is vital. This endeavour involves the comprehensive gathering of multifaceted information concerning the cityscape. This rich data encompasses complex details like road networks, public transportation systems, iconic landmarks, and prominent geographical features that together form the background against which outdoor trajectories unfold. The acquisition of this contextual data is essential in the analysis and modelling of outdoor mobility patterns.

Once the diverse benchmark datasets have been collected, the next critical step in their development is the establishment of ground truth information. This process often necessitates the careful annotation of specific characteristics within the trajectory data. These annotations serve as reference points and are indispensable for evaluating the performance of various algorithms in subsequent research and analysis. The annotation of trajectory datasets may involve a range of tasks, depending on the research objectives and the specific

application domain. Anomalies (outliers) in the trajectory data also require annotations (Corizzo, Ceci, and Japkowicz 2019). Identifying and labelling such anomalies is essential for developing anomaly detection algorithms, which play a crucial role in various fields, including security and fraud detection. Equally important in the benchmark development process is the definition of performance metrics. These metrics serve as objective measures to assess the accuracy, efficiency, and effectiveness of algorithms applied to the trajectory data. The choice of metrics depends on the specific research goals and the type of trajectory data under consideration. Commonly used performance metrics include trajectory similarity measures, which quantify how closely the predicted trajectories align with ground truth trajectories. Prediction accuracy is another critical metric, particularly in applications involving the forecasting of future trajectories.

The publication of these benchmark datasets, ground truth annotations, and evaluation scripts is a key step in the collaborative journey of advancing research in the field of mobility data analysis. By providing access to benchmark datasets, researchers offer a shared reference point. This enables fellow scholars to rigorously test and compare their models. It fosters an environment where results are not isolated in silos but can be measured against a common standard. Equally important are the ground truth annotations that serve as a compass, guiding researchers in navigating the complex terrain of trajectory analysis. These annotations provide crucial context, allowing algorithms to be trained and evaluated with accuracy. The availability of evaluation scripts further streamlines the research process. These scripts enable automated, consistent, and objective assessment of algorithm performance. Openly sharing benchmark datasets, ground truth annotations, and evaluation scripts with the research community catalyses innovation and progress. Providing a standardised foundation, it empowers researchers to refine experiments across diverse backgrounds and institutions. This accessibility extends beyond academia, influencing real-world applications, notably in tackling mobility challenges. Leveraging these resources, researchers can develop practical solutions, bridging the gap between academic insights and tangible improvements in daily life. This ensures advancements in mobility data analysis translate into meaningful societal benefits beyond research papers.

(2) **Standard Trajectory Preprocessing**: Standardising the preprocessing of trajectory datasets is vital to enhance data quality and comparability. This standardisation begins with data cleaning, where rules and procedures are defined for cleaning raw trajectory data (Lima and Souza 2023). This may involve outlier removal, handling missing values, and smoothing noisy trajectories. Coordinate systems and data formats should also be standardised. Specifying a common coordinate system and projection method for

geospatial data ensures consistency across datasets. Promoting the use of a standard data format (e.g. GeoJSON, CSV) for storing trajectory data simplifies data exchange and analysis. Additionally, encouraging the inclusion of metadata in datasets provides essential information about data sources, collection methods, and timestamps, further enhancing data quality.

(3) **Standard Feature Extraction**: Establishing standards for extracting spatio-temporal and kinematic features from trajectory datasets is another critical aspect of promoting consistency and interoperability in research. This standardisation begins with the creation of a catalogue of common features relevant to trajectory data. Features such as speed, acceleration, direction, or stop locations should be included. To ensure consistency, specify algorithms or methods for extracting these features from raw trajectory data. These methods should be well-documented and computationally efficient. Moreover, the feature extraction standards should be designed to be compatible with different types of trajectory data, whether collected from GPS devices, mobile apps, or sensors. Validation of feature extraction methods using benchmark datasets is encouraged. This validation process helps ensure that the feature extraction standards yield accurate and consistent results across different datasets and research projects.

(4) **Privacy, Sharing, and Use of Datasets**: Addressing the challenges associated with privacy considerations, sharing, and utilisation of trajectory datasets requires a comprehensive approach that spans technology, ethics, and regulation. This multifaceted strategy is crucial for preserving individual privacy while extracting valuable insights from these datasets. A fundamental step in safeguarding privacy involves the application of anonymisation and pseudonymization techniques. Anonymisation entails either the removal or encryption of personally identifiable information (PII) from trajectory datasets (Casabona 2017). Concurrently, pseudonymization replaces sensitive identifiers with pseudonyms, ensuring that direct identification of individuals from the data is prevented (Casabona 2017).

While accessing trajectory datasets from publicly accessible machines can be a practical strategy to mitigate the risks associated with theft or loss of personal devices, it is imperative to ensure that the chosen public machines are fortified with robust security mechanisms. Adopting a comprehensive approach that includes secure data storage, stringent access controls, and regular security assessments is paramount to safeguard against data breaches and protect the privacy of sensitive information. Ethical guidelines are essential in addressing privacy challenges associated with trajectory data. Adhering to these guidelines ensures responsible handling of data in line with privacy principles. Obtaining

informed consent, particularly in research involving human subjects, is imperative. Additionally, staying informed about relevant data privacy regulations, such as GDPR in Europe or Health Insurance Portability and Accountability Act (HIPAA) in the United States, is essential to avoid legal consequences. Public awareness and education initiatives are also crucial. These efforts inform the public about trajectory data usage and privacy protection measures, fostering trust and encouraging responsible data practices. Ultimately, promoting ethical awareness and education contributes to a more conscientious and privacy-conscious approach to handling trajectory datasets.

(5) **Uniform Standard for All Trajectory Datasets**: While standardisation is essential, it is important to recognise that not all trajectory datasets are the same. They can vary significantly in terms of sources, objectives, and characteristics. Therefore, advocating for a 'one-size-fits-all' uniform standard may not be suitable for all trajectory datasets. A more pragmatic approach is to adopt a set of core standards and guidelines that researchers can follow while allowing flexibility for dataset-specific adaptations when necessary. This balanced approach maintains consistency in core aspects of data handling while accommodating the diverse nature of mobility data, ensuring that research in the field remains adaptable and relevant to various data sources and research objectives.

6. Conclusion

This study delves into a comprehensive exploration of the multifaceted issues and concerns associated with the utilisation of spatio-temporal trajectory datasets. The insights derived from the focus group discussions carry substantial value for researchers, practitioners, and policymakers, enriching the existing knowledge repository of spatio-temporal-based mobility case studies. The identified challenges serve as invaluable guidance for researchers, empowering them to define and implement effective techniques and technologies for the construction, sharing, and utilisation of trajectory datasets. The nuanced and comprehensive understanding garnered from this study is poised to significantly contribute to the advancement and maturation of spatio-temporal-based mobility studies, fostering benefits for both academic and industry stakeholders. This research lays a solid foundation for continued progress in the dynamic field of spatio-temporal data analysis.

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The data collection for this research was conducted by Muhammad Arslan during the summer school, showcasing his contribution to the study. Furthermore, Arslan took the lead in writing the paper. Prior to submission, the paper underwent a thorough review by Christophe Cruz, ensuring its quality and adherence to scholarly standards.

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